

Q1 - 25 January - Shift 2

A body of mass 1 kg collides head on elastically with a stationary body of mass 3 kg. After collision, the smaller body reverses its direction of motion and moves with a speed of 2m/s. The initial speed of the smaller body before collision is ms^{-1} .

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Q2 - 25 January - Shift 2

A nucleus disintegrates into two smaller parts, which have their velocities in the ratio 3 : 2. The

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ratio of their nuclear sizes will be $\left(\frac{x}{3}\right)^{\frac{1}{3}}$. The value of 'x' is :

Q3 - 30 January - Shift 1

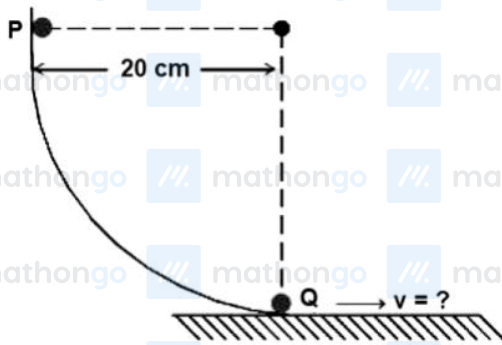
A ball of mass 200 g rests on a vertical post of height 20 m. A bullet of mass 10 g, travelling in horizontal direction, hits the centre of the ball. After collision both travels independently. The ball hits the ground at a distance 30 m and the bullet at a distance of 120 m from the foot of the post. The value of initial velocity of the bullet will be (if $g = 10 \text{ m/s}^2$) :

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- (1) 120 m/s (2) 60 m/s
(3) 400 m/s (4) 360 m/s

Q4 - 30 January - Shift 1

As per the given figure, a small ball P slides down the quadrant of a circle and hits the other ball Q of equal mass which is initially at rest. Neglecting the effect of friction and assume the collision to be elastic, the velocity of ball Q after collision will be : ($g = 10 \text{ m/s}^2$)



- (1) 0 (2) 0.25 m/s
(3) 2 m/s (4) 4 m/s

Q5 - 30 January - Shift 2

A machine gun of mass 10 kg fires 20 g bullets at the rate of 180 bullets per minute with a speed of 100 m s^{-1} each. The recoil velocity of the gun is :

- (1) 0.02 m/s
(2) 2.5 m/s
(3) 1.5 m/s
(4) 0.6 m/s

Q6 - 31 January - Shift 1

100 balls each of mass m moving with speed v simultaneously strike a wall normally and reflected back with same speed, in time t s. The total force exerted by the balls on the wall is

(1) $\frac{100mv}{t}$

(2) $\frac{200mv}{t}$

(3) $200 mvt$

(4) $\frac{mv}{100t}$

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Q7 - 31 January - Shift 1

A solid sphere of mass 1 kg rolls without slipping on a plane surface. Its kinetic energy is 7×10^{-3} J. The speed of the centre of mass of the sphere is _____ cm s^{-1} .

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Q8 - 31 January - Shift 2

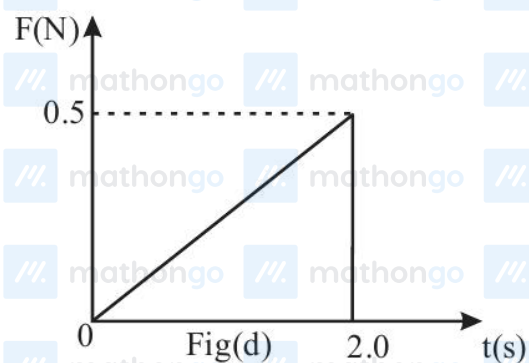
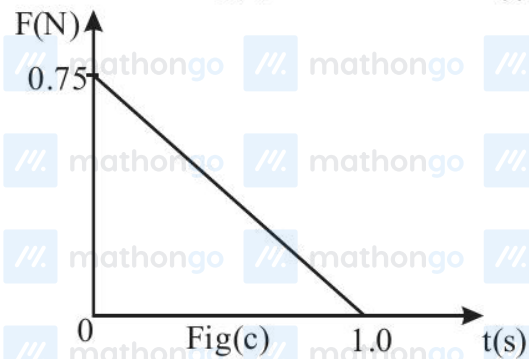
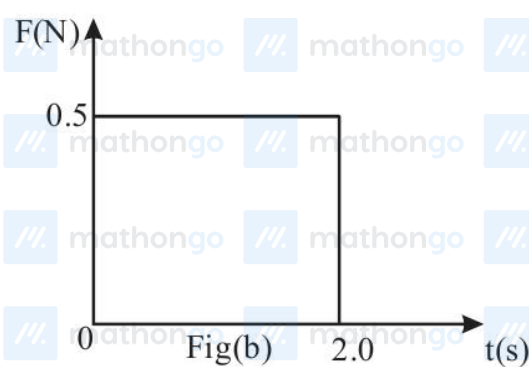
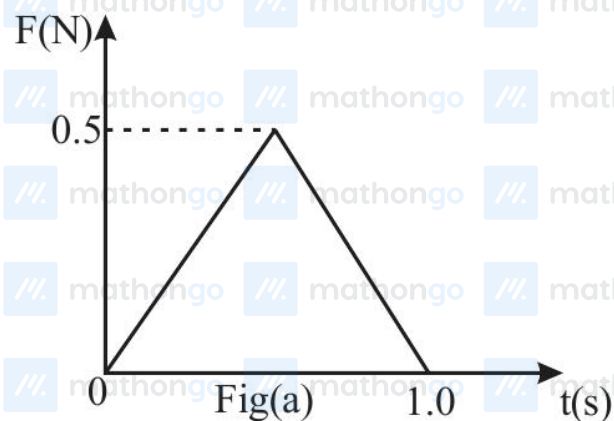
A ball is dropped from a height of 20 m. If the coefficient of restitution for the collision between ball and floor is 0.5, after hitting the floor, the ball rebounds to a height of _____ m.

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Q9 - 01 February - Shift 2

Figures (a), (b), (c) and (d) show variation of force with time.

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The impulse is highest in figure.

- (1) Fig (c) (2) Fig (b)
(3) Fig (a) (4) Fig (d)

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Answer Key

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(As per Official NTA Key released on 2 Feb)

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Q1 (4) **Q2 (2)** **Q3 (4)** **Q4 (3)**
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Q5 (4) **Q6 (2)** **Q7 (10)** **Q8 (5)**
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Q9 (2)
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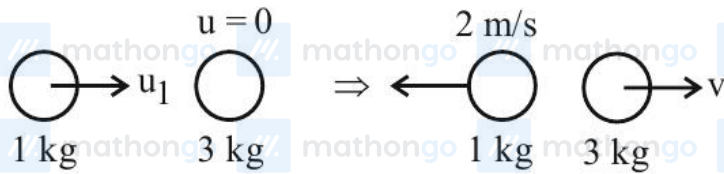
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Q1 (4)



$$1 \times u_1 = -2 + 3v \Rightarrow u_1 = -2 + 3v \dots\dots (1)$$

$$1 = \frac{v+2}{u_1} \Rightarrow v+2 = u_1 \dots\dots (2)$$

Solving (1) and (2)

$$u_1 = 4 \text{ m/s}$$

Q2 (2)



$$\frac{v_1}{v_2} = \frac{3}{2}$$

$$m_1 v_1 = m_2 v_2 \Rightarrow \frac{m_1}{m_2} = \frac{2}{3}$$

Since, Nuclear mass density is constant

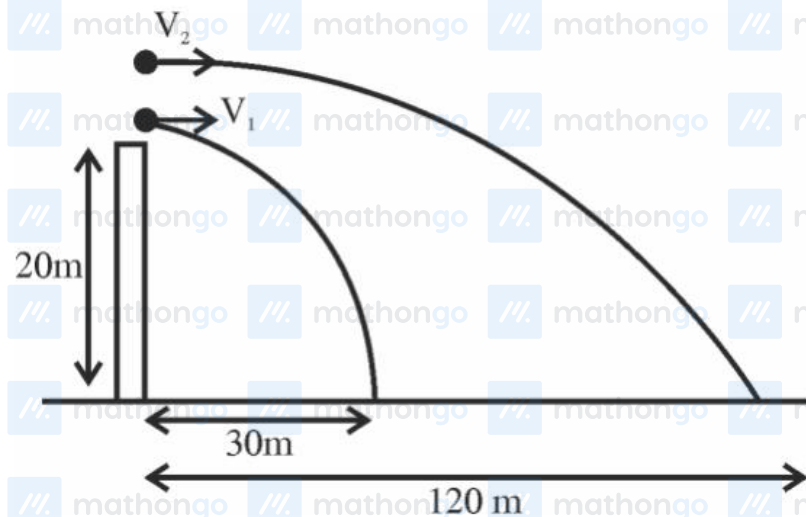
$$\frac{m_1}{\frac{4}{3}\pi r_1^3} = \frac{m_2}{\frac{4}{3}\pi r_2^3}$$

$$\left(\frac{r_1}{r_2}\right)^3 = \frac{m_1}{m_2}$$

$$\frac{r_1}{r_2} = \left(\frac{2}{3}\right)^{\frac{1}{3}}$$

So, $x = 2$

Q3 (4)



$$V_1 = \frac{30}{\sqrt{\frac{2h}{g}}}, \quad V_2 = \frac{120}{\sqrt{\frac{2h}{g}}}$$

$$(0.01)u = (0.2)\frac{30\sqrt{g}}{\sqrt{2h}} + (0.01)\frac{120\sqrt{g}}{\sqrt{2h}}$$

$$u = 300 + 60 = 360 \text{ ms}^{-1}$$

Q4 (3)

The velocities will be interchanged after collision.

Speed of P just before collision = $\sqrt{2gh}$

$$= \sqrt{2 \times 10 \times 0.2} = 2 \text{ m/s}$$

Q5 (4)

$$20 \times 10^{-3} \times \frac{180}{60} \times 100 = 10 V$$

$$\Rightarrow v = 0.6 \text{ m/s}$$

Q6 (2)

$$\vec{P}_i = Nmv \hat{i} \quad \vec{P}_f = -Nmv \hat{i}$$

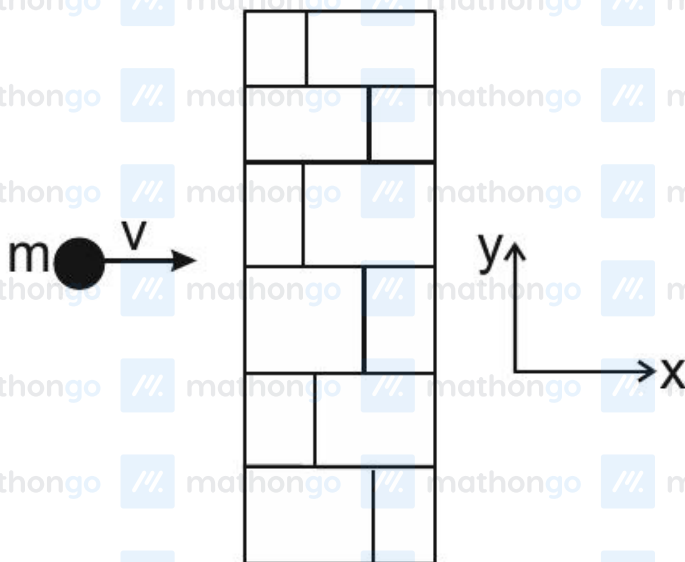
N is Number of balls strikes with wall

$$N = 100$$

$$\begin{aligned} \Delta \vec{P} &= \vec{P}_f - \vec{P}_i = -2Nmv \hat{i} \\ &= -200 Nmv \hat{i} \end{aligned}$$

$$\vec{F}_{\text{Total}} = \frac{\Delta \vec{P}}{\Delta t} = -\frac{200mvt}{t}$$

$$|\vec{F}| = \frac{200mv}{t}$$



Q7 (10)

$$\frac{1}{2}mv^2 + \frac{1}{2}I\omega^2 = 7 \times 10^{-3}$$

$$\frac{1}{2}mv^2 + \frac{1}{2}\left(\frac{2}{5}MR^2\right)\left(\frac{V}{R}\right)^2 = 7 \times 10^{-3}$$

$$\frac{1}{2}MV^2\left[1 + \frac{2}{5}\right] = 7 \times 10^{-3}$$

$$\frac{1}{2}(1)(V^2)\left(\frac{7}{5}\right) = 7 \times 10^{-3}$$

$$V^2 = 10^{-2}$$

$$V = 10^{-1} = 0.1 \text{ m/s} = 10 \text{ cm/s}$$

Ans : 10

Q8 (5)

No Solution

Q9 (2)

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Impulse = Area under $F = t$ curve

(a) $\frac{1}{2} \times 1 \times 0.5 = \frac{1}{4} \text{ N.s}$

(b) $0.5 \times 2 = 1 \text{ N.s}$ (maximum)

(c) $\frac{1}{2} \times 1 \times 0.75 = \frac{3}{8} \text{ N.s}$

(d) $\frac{1}{2} \times 2 \times 0.5 = \frac{1}{2} \text{ N.s}$